DSC5001: Statistical Machine Learning I

Week 3: Statistical Inference II: Bootstrap and Statistical Uncertainty

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Learning objectives

1. Explain a sampling distribution and standard error.
2. Implement nonparametric bootstrap resampling.
3. Compute bootstrap bias, standard error, and a percentile interval.
4. State what bootstrap uncertainty can and cannot show.

Why Estimates Vary

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The Same Estimate Can Mean Different Uncertainty

Suppose two independent studies estimate the same Bernoulli probability.

	Study A	Study B
Sample size	$n = 10$	$n = 100$
Heads	6	60
MLE	$\widehat{p} = 0.6$	$\widehat{p} = 0.6$

Both studies give exactly the same point estimate.
But the uncertainty is not the same.
For a Bernoulli proportion:

$$Var(\widehat{p}) = \frac{p(1-p)}{n}.$$

Therefore, larger n should produce a more stable estimate.

A point estimate tells us where. Uncertainty tells us how reliable.

Sampling Distribution: Formal Setup

Assume $X_1, \ldots, X_n$ are random observations from a population with unknown parameter θ.
An estimator is a rule applied to the random sample:

$$\widehat{\theta} = T(X_1, \ldots, X_n).$$

Because the sample can change from one study to another, the estimator can also take different values.
Repeated samples therefore produce:

$$\widehat{\theta}^{(1)}, \widehat{\theta}^{(2)}, \widehat{\theta}^{(3)}, \ldots$$

Definition

The sampling distribution is the probability distribution of the estimator across repeated samples from the same population.

The estimator $\widehat{\theta}$ varies because the sample varies.

Estimator Before and After Observing Data

Before observing the data, $X_1, \ldots, X_n$ are random variables, so $\widehat{\theta} = T(X_1, \ldots, X_n)$ is also random. After we observe one particular sample, we write the observed values as $x_1, \ldots, x_n$ and calculate:

$$\widehat{\theta} = T(x_1, \ldots, x_n).$$

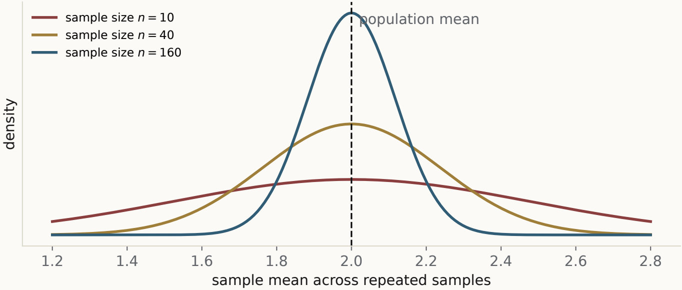
This specific numerical value is called the **estimate**.
For example, for a coin:

$$\widehat{p} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

is the estimator.
After observing H, H, T, we have $(x_1, x_2, x_3) = (1, 1, 0)$, so:

$$\widehat{p} = \frac{1 + 1 + 0}{3} = \frac{2}{3}.$$

Sample Size and the Sampling Distribution



Center
expected estimate

Width
sampling uncertainty

Larger n
narrower distribution

Increasing the sample size reduces sampling variation without changing the target parameter.

Standard Error

The variance of an estimator describes its sampling variability:

$$Var(\widehat{\theta}).$$

Its standard deviation is called the **standard error**:

$$\boxed{\mathrm{se}(\widehat{\theta}) = \sqrt{Var(\widehat{\theta})}}$$

Standard deviation and standard error answer different questions:

Standard deviation	Standard error
variation of observations	variation of an estimator
spread of X	spread of $\widehat{\theta}$

A small standard error means that repeating the study would usually produce similar estimates.

Why Does the Sample Mean Become More Stable?

Assume $X_1, \ldots, X_n$ are independent with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2$.
The sample mean is $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.
Using independence:

$$Var(\bar{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{\sigma^2}{n}.$$

Therefore:

$$\boxed{\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}}$$

Doubling the sample size does **not** halve the standard error.
To halve the standard error, we need approximately four times as many observations.

Using Standard Error to Plan Sample Size

For the sample mean:

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Suppose we want the standard error to be no larger than a target value r:

$$\frac{\sigma}{\sqrt{n}} \le r.$$

Solving for n gives:

$$\boxed{n \ge \left(\frac{\sigma}{r}\right)^2}$$

For example, if $\sigma = 10$ and we want $\mathrm{se}(\bar{X}) \le 2$:

$$n \ge \left(\frac{10}{2}\right)^2 = 25.$$

More Data Has Diminishing Returns

Because standard error decreases at rate $1/\sqrt{n}$:

Sample size	Relative standard error
n	1
$4n$	$1/2$
$9n$	$1/3$
$16n$	$1/4$

Therefore, obtaining twice the statistical precision requires approximately four times as much independent data.

With independent data from the same population, precision improves at the $1/\sqrt{n}$ rate.

Exact Sampling Distribution of the Sample Mean

Assume $X_1, \ldots, X_n$ are independent observations from $\mathcal{N}(\mu, \sigma^2)$.
A useful property of Gaussian random variables is that a sum of independent Gaussian variables is still Gaussian.
Therefore:

$$\sum_{i=1}^{n} X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

The sample mean divides this sum by n:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

so:

$$\boxed{\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)}$$

This result is exact for every sample size n.

What If the Population Is Not Gaussian?

Suppose $X_1, \ldots, X_n$ are independent observations from a non-Gaussian population.
Assume they have population mean μ and finite variance σ^2.
The sample mean still satisfies:

$$E[\bar{X}] = \mu, \qquad Var(\bar{X}) = \frac{\sigma^2}{n}$$

but its exact distribution may be difficult to derive.
The **Central Limit Theorem** tells us what happens as the sample size becomes large.
It states that the standardized sample mean:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

becomes increasingly close to a standard Gaussian distribution.

Central Limit Theorem

Formally, under suitable conditions:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } n \to \infty$$

The symbol $\xrightarrow{d}$ means **convergence in distribution**: the shape of the sampling distribution becomes closer to the target distribution.
Equivalently, for sufficiently large n:

$$\boxed{\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)}$$

Gaussian population	Non-Gaussian population
Exact for any n	Approximate for sufficiently large n

How large n must be depends on the population shape; strong skewness or heavy tails may require more observations.

What Can We Learn from a Sampling Distribution?

The sampling distribution describes the behavior of an estimator across repeated studies. Three features are especially useful:

Feature	**What it tells us**
Center	whether the estimator is systematically shifted
Spread	how much the estimate varies across samples
Shape and quantiles	how uncertainty is distributed around the estimate

These features are connected to quantities we want to estimate:

$$\text{center} \to \text{bias}, \qquad \text{spread} \to \text{standard error}.$$

Quantiles of the distribution can also be used to construct uncertainty intervals.

To quantify uncertainty, we want more than one estimate; we want information about the distribution of possible estimates.

Estimating an Unknown Standard Error

For the sample mean, the theoretical standard error is $\sigma/\sqrt{n}$.
In practice, σ is usually unknown, so we replace it with the sample standard deviation s:

$$\widehat{\mathrm{se}}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

This is an example of the **plug-in principle**: replace an unknown population quantity by an estimate from the observed data.
For a Bernoulli proportion, the same idea replaces the unknown p in the theoretical standard error by $\widehat{p}$.

Even when a standard-error formula is known, unknown quantities inside that formula must usually be estimated.

When Can We Derive a Sampling Distribution Exactly?

For some estimators, the probability model allows us to derive the sampling distribution analytically.

Model	Estimator	Sampling distribution
Gaussian	$\bar{X}$	$\mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
Bernoulli	$\widehat{p}$	scaled Binomial distribution

For Bernoulli data, if m successes are observed, $\widehat{p} = m/n$.
Because the number of successes follows a Binomial distribution, we can determine the probability of every possible value of $\widehat{p}$.
For example, when $p = 0.6$ and $n = 10$:

$$P(\widehat{p} = 0.7) = \binom{10}{7}(0.6)^7(0.4)^3 \approx 0.215.$$

When the Sampling Distribution Is Not Easy to Derive

Consider the sample $[1, 2, 4, 7, 11]$ with sample median $\widehat{\theta} = 4$.
A single estimate does not tell us how stable the estimator is.

If we repeated the study, how much would the sample median change?

Answering this requires the sampling distribution of the median.
If the population distribution F were known, we could repeatedly sample from F and recompute the median.
But in practice:

F is unknown, and we observe only one sample.

We need to estimate repeated-sampling uncertainty using only the observed data.

If the Population Distribution Were Known

Let F denote the true population distribution.
If F were known, estimating uncertainty would be straightforward.
We could repeatedly generate new datasets of size n from F:

$$D^{(1)}, D^{(2)}, \dots, D^{(B)} \sim F$$

and recompute the estimator:

$$\widehat{\theta}^{(b)} = T(D^{(b)}), \qquad b = 1, \dots, B.$$

The distribution of these values would approximate the true sampling distribution of $\widehat{\theta}$.

$$F \longrightarrow \text{repeated datasets} \longrightarrow \text{repeated estimates} \longrightarrow \text{sampling distribution}$$

But the Population Distribution Is Unknown

The true population distribution F is usually unknown.
What we actually observe is:

$$x_1, x_2, \ldots, x_n.$$

The **empirical distribution** $\widehat{F}_n$ uses these observations as an estimate of F.
It assigns probability $1/n$ to each observed data point.
For example, for $[2, 4, 9, 10]$:

Observed value	2	4	9	10
Probability under $\widehat{F}_n$	$1/4$	$1/4$	$1/4$	$1/4$

$$\boxed{F \text{ unknown} \quad \longrightarrow \quad \widehat{F}_n \text{ estimated from data}}$$

The empirical distribution is our data-based approximation to the unknown population distribution.

From the Plug-In Principle to Resampling

Earlier, when σ was unknown, we replaced it with the sample estimate s.
The same **plug-in principle** can be applied to the entire population distribution:

$$F \quad \longrightarrow \quad \widehat{F}_n.$$

If we cannot draw new samples from the unknown F, we can draw samples from $\widehat{F}_n$ instead.
A new simulated sample would therefore contain observations selected from:

$$x_1, x_2, \ldots, x_n.$$

Because each draw comes independently from $\widehat{F}_n$, an observation may appear more than once, while another may not appear at all.

Can repeated samples from $\widehat{F}_n$ approximate repeated samples from F?

This replacement of F by $\widehat{F}_n$ is the central idea that leads to the bootstrap.

Bootstrap Principle

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From the Empirical Distribution to Bootstrap Samples

From the previous section, the unknown population distribution F is replaced by the empirical distribution $\widehat{F}_n$.
The bootstrap now treats $\widehat{F}_n$ as if it were the population.
For bootstrap repetition b, draw n observations independently from $\widehat{F}_n$:

$$X_1^{*(b)}, \dots, X_n^{*(b)} \overset{i.i.d.}{\sim} \widehat{F}_n.$$

These observations form the bootstrap sample $D^{*(b)}$.
Apply exactly the same estimator T:

$$\widehat{\theta}^{*(b)} = T(D^{*(b)}).$$

Definition

A bootstrap replicate is the estimate obtained by applying the original estimator to one bootstrap resample.

Why Must Bootstrap Sample With Replacement?

Consider the observed sample:

$$D = [1, 2, 4, 7, 11].$$

If we sample five observations **without replacement**, every resample contains exactly the same values, only in a different order:

$$[7, 1, 11, 2, 4].$$

For statistics such as the mean or median, this gives exactly the same estimate.
There is no resampling variation.
With replacement, samples such as:

$$[1, 1, 2, 7, 11] \qquad \text{or} \qquad [2, 4, 7, 7, 11]$$

become possible.

Sampling with replacement creates variation among resamples while preserving the empirical distribution as the sampling source.

Bootstrap Sample Coverage

For any observation in a dataset of size n, the probability that it is never selected in n bootstrap draws is:

$$P(\text{not selected}) = \left(1 - \frac{1}{n}\right)^n \longrightarrow e^{-1} \approx 0.368.$$

Therefore:

$$P(\text{selected at least once}) \approx 1 - e^{-1} \approx 0.632.$$

Hence, the expected number of distinct observations is:

$$\boxed{E[\text{unique observations}] \approx 0.632n}$$

A bootstrap sample contains n draws, but only about 63.2% distinct original observations on average.

Why Does Each Bootstrap Sample Still Have Size n?

The original estimator was computed from a sample of size n.
The bootstrap tries to reproduce the uncertainty of that same experiment.
If we instead resampled only m observations, we would approximate the uncertainty of an experiment with sample size m.
For many estimators:

$$\mathrm{se}(\widehat{\theta}) \propto \frac{1}{\sqrt{n}}.$$

Changing the resample size therefore changes the uncertainty we are trying to estimate.

A Bootstrap Example: Estimating a Median

Consider the observed sample:

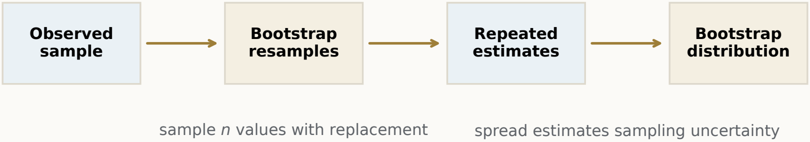
$$D = [1, 2, 4, 7, 11].$$

The statistic of interest is the median, so the original estimate is $\widehat{\theta} = 4$.
Possible bootstrap samples include:

Bootstrap sample	**Bootstrap median**
$[1, 1, 2, 7, 11]$	2
$[1, 4, 4, 7, 11]$	4
$[2, 4, 7, 7, 11]$	7

Every resample contains five observations, but some observations are repeated and others are omitted.
The statistic therefore changes across bootstrap samples.

Bootstrap Workflow



The same estimator T is applied throughout:

$$\widehat{\theta} = T(D), \qquad \widehat{\theta}^{*(b)} = T(D^{*(b)}).$$

Examples of T include the sample mean, median, or MLE.

Bootstrap changes the data through resampling, not the estimation rule.

What Is Bootstrap Approximating?

In the real sampling world:

$$F \longrightarrow D^{(b)} \longrightarrow \widehat{\theta}^{(b)}.$$

If we could repeat this experiment, we could observe how $\widehat{\theta}$ varies around the true parameter θ.
In the bootstrap world:

$$\widehat{F}_n \longrightarrow D^{*(b)} \longrightarrow \widehat{\theta}^{*(b)}.$$

We instead observe how the bootstrap estimates vary around the original estimate $\widehat{\theta}$.
The central approximation is:

$$\boxed{\widehat{\theta}^* - \widehat{\theta} \quad \text{mimics} \quad \widehat{\theta} - \theta}$$

Bootstrap uses variation around the observed estimate to approximate variation around the unknown true parameter.

What Is the Bootstrap Distribution?

After B bootstrap repetitions, we obtain:

$$\widehat{\theta}^{*(1)}, \ldots, \widehat{\theta}^{*(B)}.$$

The empirical distribution of these bootstrap replicates is called the **bootstrap distribution**. It approximates the sampling behavior of the original estimator $\widehat{\theta}$.

Sampling distribution	Bootstrap distribution
Repeated samples from F	Resamples from $\widehat{F}_n$
Usually unavailable	Computed from observed data

Common mistake

The bootstrap distribution is not the true sampling distribution; it is a data-based approximation.

Bootstrap Standard Error

The spread of the bootstrap replicates estimates the spread of the unknown sampling distribution.
First compute their mean:

$$\overline{\widehat{\theta}^*} = \frac{1}{B}\sum_{b=1}^{B}\widehat{\theta}^{*(b)}.$$

Then compute their standard deviation:

$$\boxed{\widehat{\mathrm{se}}_{\mathrm{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\theta}^{*(b)} - \overline{\widehat{\theta}^*}\right)^2}}$$

The $B-1$ denominator appears because we are calculating the sample standard deviation of the B bootstrap replicates.

Why Can Bootstrap Estimate Bias?

The true bias of an estimator is:

$$\mathrm{Bias}(\widehat{\theta}) = E_F[\widehat{\theta}] - \theta.$$

But both the population distribution F and the true parameter θ are unknown. Bootstrap replaces them by:

$$F \to \widehat{F}_n, \qquad \theta \to \widehat{\theta}.$$

Therefore, in the bootstrap world:

$$\boxed{\widehat{\mathrm{Bias}}_{\mathrm{boot}} = \overline{\widehat{\theta}^*} - \widehat{\theta}}$$

Bootstrap estimates the unknown center shift by reproducing the same comparison inside the empirical distribution.

Interpreting and Correcting Bootstrap Bias

Suppose:

$$\widehat{\theta} = 10, \qquad \overline{\widehat{\theta}^*} = 10.8.$$

Then:

$$\widehat{\text{Bias}}_{\text{boot}} = 10.8 - 10 = 0.8.$$

The bootstrap suggests that the estimator is systematically shifted upward.
A simple bias correction subtracts this estimated shift:

$$\widehat{\theta}_{\text{corrected}} = \widehat{\theta} - \widehat{\text{Bias}}_{\text{boot}}.$$

Therefore:

$$\boxed{\widehat{\theta}_{\text{corrected}} = 10 - 0.8 = 9.2}$$

Is the Estimated Bias Large?

Bootstrap bias measures the shift of the bootstrap distribution's center:

$$\widehat{\text{Bias}}_{\text{boot}} = \overline{\widehat{\theta}^*} - \widehat{\theta}.$$

But the same bias value can mean different things for estimators with different sampling variability.
One simple way to interpret the bias is to compare it with the bootstrap standard error:

$$\boxed{R_{\text{bias}} = \frac{\left|\widehat{\text{Bias}}_{\text{boot}}\right|}{\widehat{\text{se}}_{\text{boot}}}}$$

A small ratio means the center shift is small relative to ordinary sampling variation.
A large ratio means the systematic shift is substantial relative to the spread.

The magnitude of bias should be interpreted relative to the estimator's sampling variability.

Worked Example: Bootstrap Median

Suppose the original median is $\widehat{\theta} = 4$.
Eight bootstrap repetitions give:

$$2, \quad 4, \quad 4, \quad 7, \quad 4, \quad 2, \quad 7, \quad 4.$$

Their mean is $\overline{\widehat{\theta}^*} = 4.25$, so:

$$\widehat{\text{Bias}}_{\text{boot}} = 4.25 - 4 = 0.25.$$

The bootstrap standard error is:

$$\widehat{\text{se}}_{\text{boot}} \approx 1.91.$$

Therefore:

$$\boxed{R_{\text{bias}} = \frac{0.25}{1.91} \approx 0.13}$$

The estimated center shift is small relative to the sampling variability.

Checking the Bootstrap Against a Known Result

For the sample mean, the analytic standard error is known:

$$\widehat{\mathrm{se}}_{\mathrm{analytic}}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

We can also estimate the same uncertainty using bootstrap resampling.
If the bootstrap is working well, the two estimates should be reasonably close.
A large disagreement suggests checking:

- whether B is large enough;
- whether the original sample is too small or unusual;
- whether independence is reasonable;
- whether the implementation is correct.

An estimator with a known analytic standard error provides a useful sanity check for bootstrap implementation.

Bootstrap Variance of the Sample Mean

For the observed sample, define:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Under the empirical distribution $\widehat{F}_n$, each observed value has probability $1/n$. Therefore:

$$E_*(X^*) = \bar{x}, \qquad Var_*(X^*) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}s^2.$$

A bootstrap mean averages n independent draws from $\widehat{F}_n$:

$$\bar{X}^* = \frac{1}{n}\sum_{i=1}^{n} X_i^*.$$

Hence:

$$\boxed{Var_*(\bar{X}^*) = \frac{n-1}{n^2}s^2}$$

Why Does This Result Matter?

For the sample mean, the familiar estimated standard error is:

$$\widehat{\text{se}}_{\text{analytic}} = \frac{s}{\sqrt{n}}.$$

From the bootstrap calculation:

$$\widehat{\text{se}}_{\text{boot}} = \sqrt{Var_*(\bar{X}^*)} = \frac{s}{\sqrt{n}}\sqrt{\frac{n-1}{n}}.$$

Compare the two:

$$\boxed{\frac{\widehat{\text{se}}_{\text{boot}}}{\widehat{\text{se}}_{\text{analytic}}} = \sqrt{\frac{n-1}{n}} \longrightarrow 1}$$

For example:

$$n = 10:\ 0.949, \qquad n = 50:\ 0.990, \qquad n = 100:\ 0.995.$$

For the sample mean, bootstrap resampling recovers almost the same sampling uncertainty as the analytic SE, using only the observed data.

Where Does Bootstrap Approximation Error Come From?

Bootstrap uses the observed sample in two steps.
Step 1: Approximate the unknown population

$$F \longrightarrow \widehat{F}_n.$$

This approximation depends on the original data.

- Better or more representative data improve this step.
- Increasing B cannot fix a poor original sample.

Step 2: Approximate the bootstrap distribution

$$\widehat{F}_n \longrightarrow B \text{ bootstrap resamples.}$$

This approximation depends on the number of repetitions B.

- Larger B reduces simulation noise.
- It does not add new population information.

Bootstrap Inference and Reliability

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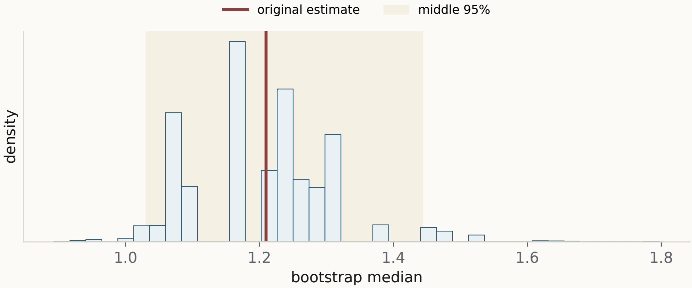
Reading a Bootstrap Distribution



Center
bias

Spread
standard error

Quantiles
uncertainty interval

A bootstrap distribution provides information about center, spread, and shape.

Percentile Bootstrap Interval

Let $\widehat{\theta}^*_\alpha$ denote the α quantile of the bootstrap replicates.
For a 95% interval:

$$\boxed{CI^{\text{perc}}_{0.95} = \left[\widehat{\theta}^*_{0.025}, \widehat{\theta}^*_{0.975}\right]}$$

bootstrap replicates $\longrightarrow$ sort $\longrightarrow$ 2.5% and 97.5% quantiles

Unlike $\widehat{\theta} \pm 1.96\,\widehat{\mathrm{se}}$, the interval need not be symmetric.

The percentile interval uses quantiles of the full bootstrap distribution.

Worked Example: Percentile Interval

Suppose $B = 2000$ bootstrap repetitions give:

Quantile	2.5%	50%	97.5%
Bootstrap median	3.1	4.4	6.8

Therefore:

$$\boxed{CI^{\text{perc}}_{0.95} = [3.1,\ 6.8]}$$

Relative to the bootstrap median:

$$4.4 - 3.1 = 1.3, \qquad 6.8 - 4.4 = 2.4.$$

The bootstrap distribution is asymmetric.

What Does 95% Confidence Mean?

Let θ be the fixed but unknown population parameter.
Repeated studies produce different datasets and different intervals:

$$D^{(1)} \to C^{(1)}, \qquad D^{(2)} \to C^{(2)}, \qquad D^{(3)} \to C^{(3)}, \qquad \dots$$

The true parameter θ is the same in every study.
A 95% confidence procedure aims for:

$$\boxed{P_F(\theta \in C(D)) \approx 0.95}$$

In repeated studies, about 95% of the intervals would contain the same true θ.

Common mistake

The interval changes across samples; the true parameter θ does not.

Checking Confidence-Interval Coverage

In a simulation, suppose the true F and θ are known.
Repeat the complete experiment M times:

$$D^{(m)} \longrightarrow C^{(m)}, \qquad m = 1, \ldots, M.$$

For each interval, record whether it contains θ:

$$I_m = \begin{cases} 1, & \theta \in C^{(m)}, \\ 0, & \theta \notin C^{(m)}. \end{cases}$$

The empirical coverage is:

$$\boxed{\widehat{\text{Coverage}} = \frac{1}{M} \sum_{m=1}^{M} I_m}$$

For a good 95% interval procedure:

$$\widehat{\text{Coverage}} \approx 0.95.$$

SE-Based vs. Percentile Intervals

	SE-based	**Percentile**
Interval	$\widehat{\theta} \pm 1.96\,\widehat{\mathrm{se}}$	$[\widehat{\theta}^*_{0.025}, \widehat{\theta}^*_{0.975}]$
Shape	symmetric	can be asymmetric
Uses	center + spread	bootstrap quantiles
Natural when	near-Gaussian	bootstrap shape is informative

Two distributions may have the same SE but different shapes:

$$\text{symmetric} \qquad \neq \qquad \text{right-skewed.}$$

When Percentile Intervals Can Fail

Percentile intervals may have poor coverage when the bootstrap distribution does not reproduce the true sampling distribution well.

Issue	Possible consequence
substantial estimator bias	shifted interval
very small n	poor empirical approximation
unstable statistic	irregular bootstrap distribution
poor sampling design	wrong target population

A more sophisticated interval cannot repair unrepresentative data.

How Large Should B Be?

B is the number of bootstrap repetitions.
Using the same original dataset, suppose we obtain:

B	Bootstrap SE	95% percentile interval
500	1.88	$[2.9, 6.9]$
2000	1.91	$[3.1, 6.8]$
10000	1.90	$[3.1, 6.8]$

As B increases, the results become stable.

$$\boxed{\text{Little change when } B \text{ increases} \quad \Rightarrow \quad B \text{ is large enough}}$$

Failure Case 1: Unrepresentative Data

Target population:

$$F = \text{all adults in a city.}$$

Observed sample:

$$D = \text{university students only.}$$

Then:

$$F \not\approx \widehat{F}_n \quad \Longrightarrow \quad D^* \sim \widehat{F}_n \text{ remains unrepresentative.}$$

Even if:

$$B \to \infty,$$

the missing population groups do not appear.

Bootstrap reproduces the observed data; it does not repair sampling bias.

Failure Case 2: Dependence

Ordinary bootstrap assumes that the resampled units represent independent information.

Data structure	**Resample**
independent observations	individual observations
repeated patient measurements	patients
clustered observations	clusters
time series	consecutive blocks

Example:

$$100 \text{ patients} \times 5 \text{ measurements} = 500 \text{ rows},$$

but the natural independent unit may still be the patient.

Resample the independent unit, not automatically each row.

Failure Case 3: Rare Events

Suppose:

$$D = [0, 0, 0, 0, 0].$$

Then the empirical distribution satisfies:

$$\widehat{F}_n(0) = 1.$$

Therefore every nonparametric bootstrap sample gives:

$$D^* = [0, 0, 0, 0, 0], \qquad \widehat{p}^* = 0.$$

Hence:

$$\mathrm{Var}_{\mathrm{boot}}(\widehat{p}^*) = 0.$$

But this does not imply that the true population probability is exactly zero.

Bootstrap cannot reproduce events that never appear in the observed sample.

Failure Case 4: Sample Maximum

Suppose:

$$X_1, \dots, X_n \overset{i.i.d.}{\sim} \text{Uniform}(0, \theta), \qquad \widehat{\theta} = \max(X_1, \dots, X_n).$$

A bootstrap sample contains only observed values, so:

$$\boxed{\widehat{\theta}^* \leq \widehat{\theta}}$$

If the observed maximum is unique:

$$P_*(\widehat{\theta}^* = \widehat{\theta}) = 1 - \left(1 - \frac{1}{n}\right)^n \approx 0.632.$$

Thus about 63.2% of bootstrap samples have exactly the same maximum.

Bootstrap cannot reproduce values beyond the observed maximum, so it may fail for boundary or extreme-value estimators.

Two Ways to Generate Bootstrap Samples

So far, we have resampled directly from the observed data:

$$D = [x_1, \ldots, x_n] \longrightarrow \widehat{F}_n \longrightarrow D^*.$$

A bootstrap sample therefore contains only observed values.
An alternative is to first fit a probability model:

$$D \longrightarrow F_{\widehat{\theta}} \longrightarrow D^*.$$

For example, after fitting a Gaussian model:

$$x_i^* \sim \mathcal{N}(\widehat{\mu}, \widehat{\sigma}^2).$$

Nonparametric vs. Parametric Bootstrap

The two procedures have different names:

	Nonparametric	**Parametric**
Resample from	$\widehat{F}_n$	$F_{\widehat{\theta}}$
New values	observed values only	values generated by the model
Requires	representative sample	appropriate model

For a Gaussian model:

$$\text{data} \to (\widehat{\mu}, \widehat{\sigma}^2) \to x_i^* \sim \mathcal{N}(\widehat{\mu}, \widehat{\sigma}^2).$$

Nonparametric bootstrap trusts the observed data; parametric bootstrap also trusts a chosen probability model.

What Should Be Resampled?

Bootstrap should mimic what would be independently sampled if the study were repeated. The **resampling unit** is the basic unit drawn with replacement.

Data setting	**Resampling unit**
Independent test examples	individual examples
Repeated measurements from patients	patients
Students grouped within schools	schools or other independent clusters
Time-series observations	consecutive time blocks

For example, five measurements from one patient are not five independent patients.

Example: Comparing Two Classifiers

Suppose two classifiers are evaluated on the same n test examples.
For test example i, define:

$$C_{Ai} = \begin{cases} 1, & \text{Model A is correct,} \\ 0, & \text{otherwise,} \end{cases} \qquad C_{Bi} = \begin{cases} 1, & \text{Model B is correct,} \\ 0, & \text{otherwise.} \end{cases}$$

Their test accuracies are:

$$\widehat{\mathrm{Acc}}_A = \frac{1}{n}\sum_{i=1}^{n} C_{Ai}, \qquad \widehat{\mathrm{Acc}}_B = \frac{1}{n}\sum_{i=1}^{n} C_{Bi}.$$

Suppose:

$$\widehat{\mathrm{Acc}}_A = 0.87, \qquad \widehat{\mathrm{Acc}}_B = 0.84.$$

$$\boxed{\widehat{\Delta} = 0.87 - 0.84 = 0.03}$$

Is Model A really better, or could the 3% difference be caused by the particular test sample?

Paired Bootstrap for Model Comparison

Both models were tested on the same examples, so each test example forms a pair:

$$(C_{Ai}, C_{Bi}).$$

For each bootstrap repetition:

1. Resample n test-example indices with replacement.
2. Keep Models A and B together for each selected example.
3. Recompute both accuracies and their difference.

For repetition b:

$$\widehat{\Delta}^{*(b)} = \widehat{\mathrm{Acc}}_A^{*(b)} - \widehat{\mathrm{Acc}}_B^{*(b)}.$$

After B repetitions:

$$\widehat{\Delta}^{*(1)}, \ldots, \widehat{\Delta}^{*(B)}$$

gives the bootstrap distribution of the accuracy difference.

Resample test examples, not the two models separately, because their predictions are paired on the same examples.

Bootstrap and Bagging

Bagging means **bootstrap aggregating**: train a model on each bootstrap sample and combine the predictions.
Both methods begin with:

$$D \longrightarrow D^{*(1)}, D^{*(2)}, \ldots, D^{*(B)}.$$

What happens next is different:

	Bootstrap inference	**Bagging**
For each sample	compute a statistic	train a model
Use replicates to	study variation	make predictions
Combine by	distribution / quantiles	averaging / voting
Goal	quantify uncertainty	improve prediction stability

The resampling mechanism is similar, but bootstrap inference and bagging have different goals.

Bootstrap Reliability Checklist

Before interpreting a bootstrap result, check:

Issue	**Question**	**If not**
Population	Does the sample represent the target population?	More resampling will not fix it
Dependence	Is the correct independent unit resampled?	Use cluster / block resampling
Estimator	Is the statistic reasonably stable?	Bootstrap distribution may be unreliable
Simulation	Are results stable as B increases?	Increase B

A large B fixes simulation noise only; it cannot fix poor data, dependence, or an unstable estimator.

Bootstrap and Uncertainty in Modern AI

DSC5001 ▪ Semester A, 2026/27

Why Modern AI Scores Need Uncertainty

Modern AI systems are evaluated on finite test sets:

$$D_{\mathrm{eval}} = \{z_1, \ldots, z_n\}.$$

A reported score is therefore an estimate:

$$\widehat{S} = T(D_{\mathrm{eval}}).$$

A different test sample can give a different score.
Bootstrap asks:

$$D_{\mathrm{eval}} \longrightarrow D^*_{\mathrm{eval}} \longrightarrow \widehat{S}^* \longrightarrow \text{uncertainty.}$$

Point estimate	Uncertainty
How well did the model score?	How stable is that score?

Benchmark performance is an estimate based on a finite evaluation sample.

Example 1: Frontier LLM Evaluation

An ICLR 2026 study evaluated frontier LLMs on real forecasting events.

Model	Brier score (95% CI)
GPT-5	0.184 ± 0.006
Grok 4	0.189 ± 0.005
Claude Sonnet 4	0.194 ± 0.006
Gemini 2.5 Flash	0.197 ± 0.007

Lower Brier score is better.
The confidence intervals were obtained by bootstrap resampling evaluation events.

$$\boxed{\text{score} + \text{uncertainty} > \text{score alone}}$$

Source: *LLM-as-a-Prophet: Understanding Predictive Intelligence with Prophet Arena*, ICLR 2026.

Example 2: Computer Vision

Consider a segmentation model evaluated on n test images.
For image i, let:

$$D_i = \mathrm{Dice}(\widehat{M}_i, M_i),$$

where $\widehat{M}_i$ is the predicted mask and M_i is the ground truth.
The average test score is:

$$\widehat{D} = \frac{1}{n}\sum_{i=1}^{n} D_i.$$

Image-level bootstrap gives:

$$\{1, \dots, n\} \longrightarrow \{i_1^*, \dots, i_n^*\} \longrightarrow \widehat{D}^*.$$

Repeating this produces uncertainty for the reported segmentation score.

For vision benchmarks, resample test images rather than individual pixels.

Modern vision foundation models include DINOv3 and SAM 2.

Why This Matters for Vision Foundation Models

Modern vision models are evaluated across different datasets and domains.
For example:

$$\text{natural images} \to \text{urban scenes} \to \text{medical images} \to \text{satellite images.}$$

A performance difference may come from:

$$\boxed{\text{model difference} + \text{finite test sample} + \text{domain shift}}$$

Recent vision-foundation-model benchmarks show that model rankings can change across datasets and domain shifts.
Examples: DINOv3; SAM 2; vision-foundation-model segmentation benchmarks.

Example 3: Medical Imaging

In medical AI, the natural evaluation unit is often the **patient**.
For an AUROC evaluation:

$$\text{patients} \longrightarrow \text{resample patients} \longrightarrow \mathrm{AUROC}^* \longrightarrow 95\% \text{ CI}.$$

A recent breast MRI model reported:

External dataset	AUROC (95% CI)
DS2	$0.899\ [0.877, 0.922]$
DS3	$0.806\ [0.790, 0.822]$

The intervals were estimated using 1000 bootstrap repetitions.

Clinical performance should include uncertainty across patients, not only a single AUROC value.

Source: breast MRI large-model study, *Nature Communications*, 2025.

Medical AI: What Should Be Resampled?

Suppose one patient contributes several images:

$$\text{Patient 1} \to I_{11}, I_{12}, I_{13}.$$

These images are not independent patients.
Therefore:

Incorrect	**Better**
resample individual images	resample patients
break patient dependence	keep patient data together

The same issue appears with:

multiple CT slices, video frames, repeated visits.

In medical imaging, the resampling unit should match the independent clinical unit.

Example 4: Geospatial Foundation Models

Modern remote-sensing foundation models include Prithvi-EO-2.0, Clay, TerraMind, and other GeoFMs.
They are evaluated across:

$$\text{different sensors} + \text{resolutions} + \text{regions} + \text{tasks}.$$

Prithvi-EO-2.0, for example, was benchmarked across Earth-observation tasks spanning resolutions from approximately 0.1 to 15 m.
Recent GEO-Bench evaluations also show:

$$\boxed{\text{no single GeoFM dominates every task and domain}}$$

In remote sensing, model performance depends strongly on both task and geographic domain.

Sources: Prithvi-EO-2.0; GEO-Bench-2.

Remote Sensing: Geography Changes Bootstrap

Satellite pixels and nearby image patches are spatially correlated.
Therefore:

$$\text{pixel} \neq \text{independent observation.}$$

Consider:

$$\text{Region} \to \text{scene} \to \text{patches} \to \text{pixels.}$$

If the scientific question concerns performance across geographic areas, a more appropriate bootstrap may resample:

$$\boxed{\text{regions or spatial blocks}}$$

rather than individual pixels.

Spatial dependence determines the appropriate resampling unit in remote-sensing evaluation.

Deep Learning: Ensembles and Uncertainty

Train K neural networks independently $f_1, \ldots, f_K$.
For class probability:

$$p_1(y \mid x), \ldots, p_K(y \mid x).$$

Average prediction:

$$\overline{p}(y \mid x) = \frac{1}{K} \sum_{k=1}^{K} p_k(y \mid x).$$

Variation across models:

$$\widehat{Var}[p(y \mid x)] = \frac{1}{K-1} \sum_{k=1}^{K} \left(p_k(y \mid x) - \overline{p}(y \mid x)\right)^2.$$

Common mistake

Different training seeds create an ensemble, but not necessarily a bootstrap: the training data may be unchanged.

Modern AI Has Multiple Sources of Uncertainty

Source	Example
Evaluation sample	prompts, images, patients, regions
Training data	different training samples
Optimization	initialization and stochastic training
Generation	stochastic outputs from generative models
Domain shift	new hospitals, sensors, regions, populations

Bootstrap usually targets uncertainty associated with the chosen **resampling unit**.

$\boxed{\text{More bootstrap repetitions} \neq \text{all sources of model uncertainty}}$

First identify what is random; then choose the uncertainty method.

Week 3: connect the statistical question, mathematics, and evidence

Questions and discussion